IoT Based Smart Waste Collection System

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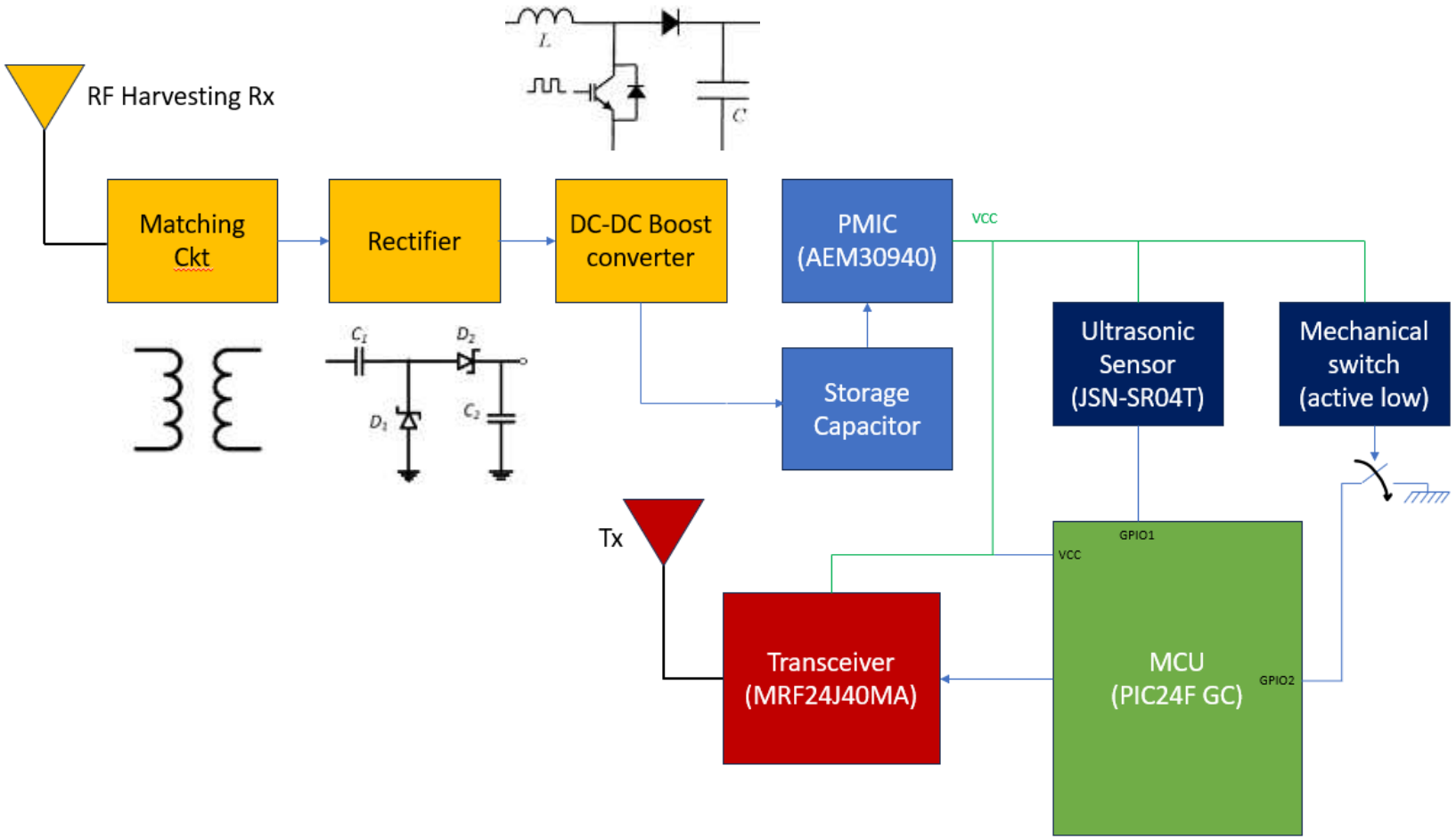


Next-gen sensing using IoT, Cloud and Battery-less SoC Design

- ❑No Batteries required
- ❑No GPS chips required
- ❑Completely automated pipeline: From bins to control room dashboard with algorithmic collection truck dispatch
- ❑On-cloud inference with Machine Learning for pattern prediction and learning
- ❑Using wasted and ambient signal power for smart waste collection!

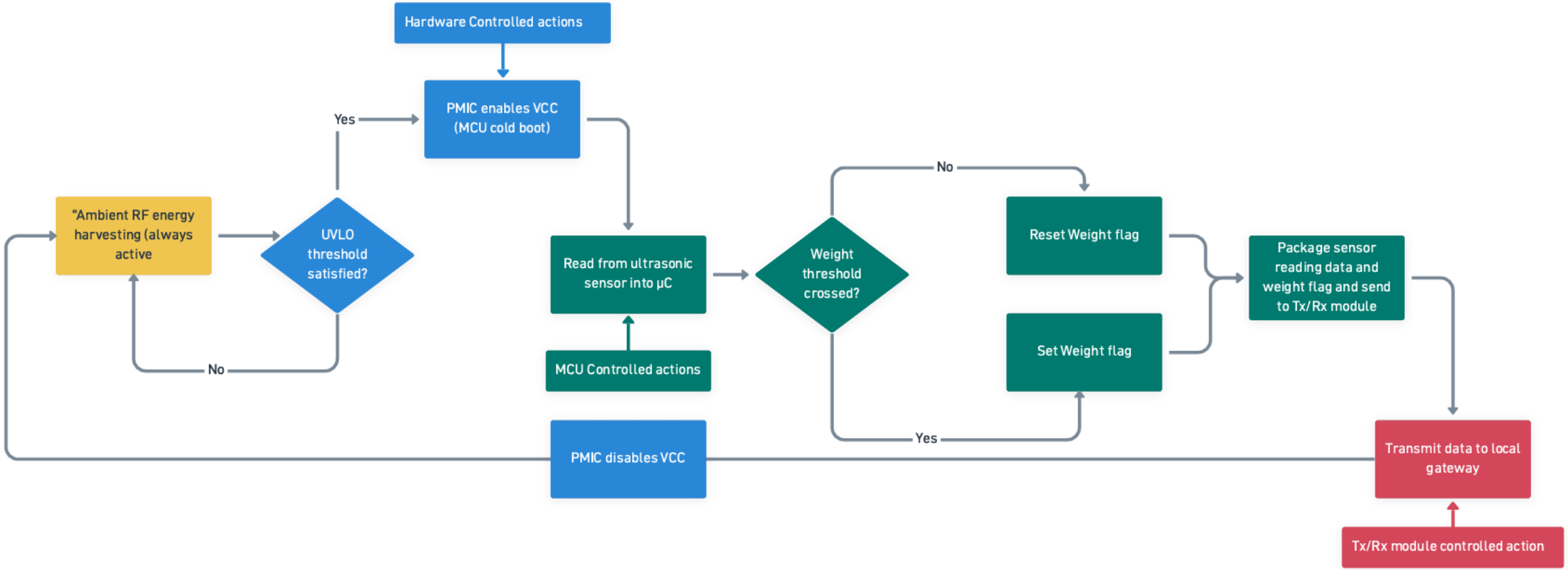
*Based on this ideation, I was competitively selected for a Project Associate position at the Smart City Research Centre, IIITH.

Sensor Node SoC Design

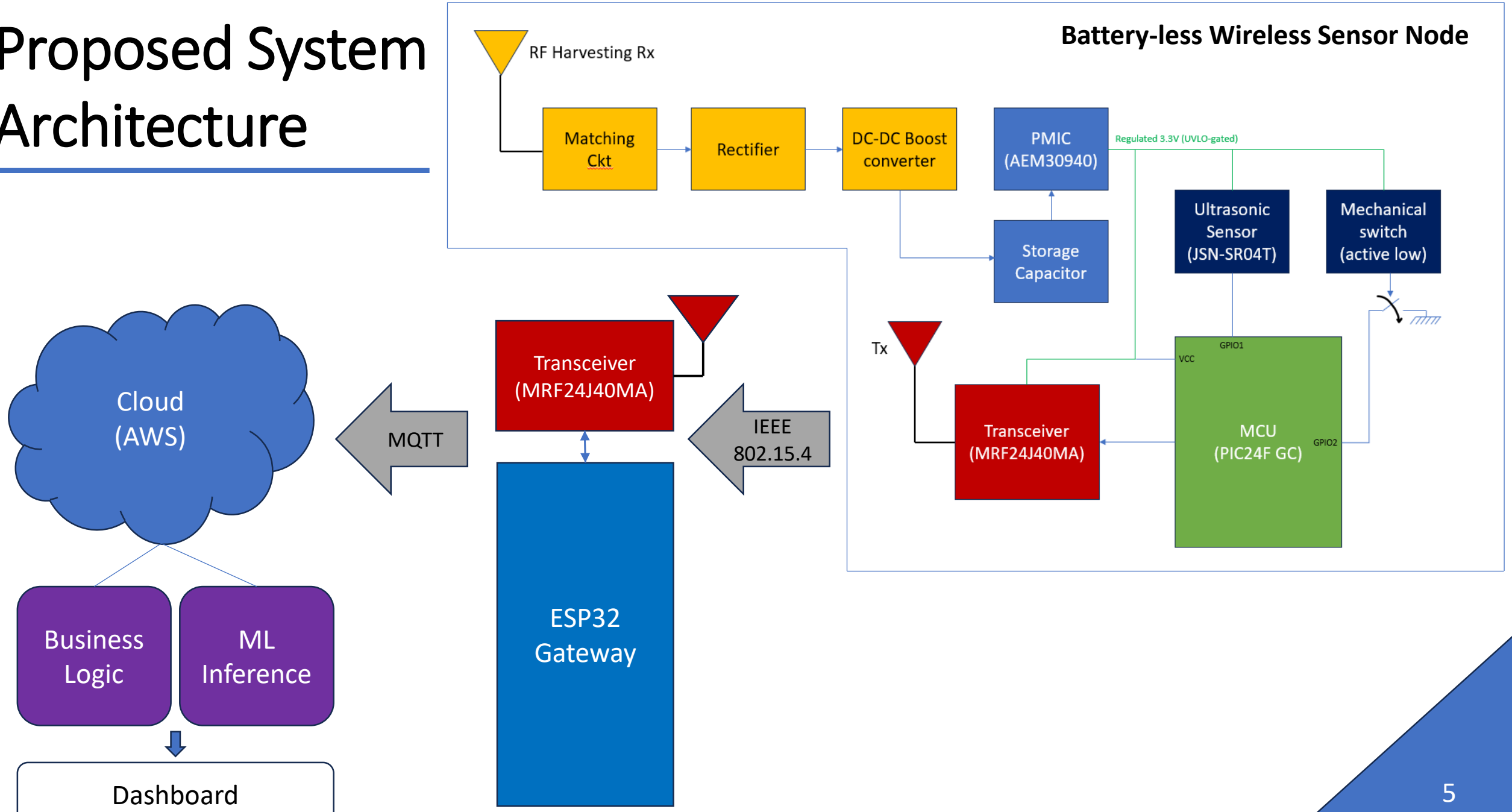


Reference: [RF energy harvesting and wireless power - EDN]

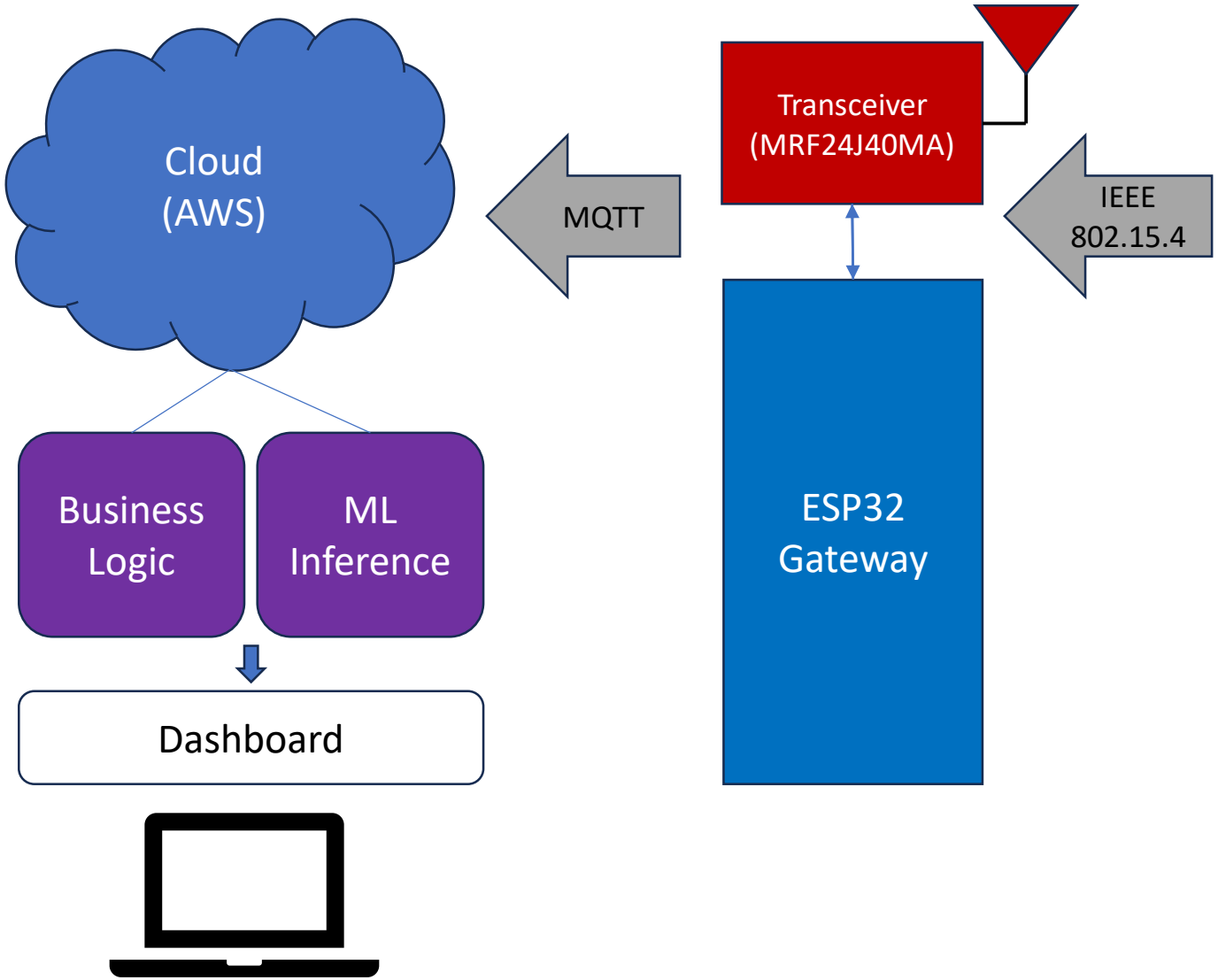
Sensor Node Operation Logic



Proposed System Architecture



Reference: [RF energy harvesting and wireless power - EDN]



Battery-less Wireless Sensor Node

Ambient RF Power Analysis

Environment	Harvested Power (after rectifier)*
Near cellular tower/outdoors	5-10 µW
Typical urban outdoor	1-3 µW
Indoor/shielded bin	0.1-1 µW
Worst case	<0.1 µW

*Values were determined via literature review[1], [2].

[1] M. Latrach, D. Bouchouicha, F. Dupont, and L. Ventura, "Ambient RF Energy Harvesting," Proc. Int. Conf. Renewable Energies Power Qual., Granada, Spain, 2010, pp. 1–6.
[2] D. Bouchouicha et al., "Measurements of ambient RF power density in urban environments," ICREPQ'10, 2010.

Estimated Energy Consumed per “wake-up”

Block	Typical Energy
PIC24 MCU (boot + logic, ~5 ms)	~0.05 mJ
Ultrasonic sensor burst	~2-3 mJ
Mechanical switch read	~0 mJ
RF TX (MRF24J40MA, ~5 ms @ 20 mA, 3.3 V)	~0.33 mJ
Regulator + losses	~0.5-1 mJ
Total per cycle	**~4-5 mJ**

- For our analysis, we shall assume an average of ~5 mJ per wake-up cycle.

Wake-up Frequency

- Wake interval = $\frac{Energy\ required\ per\ cycle}{Average\ harvested\ RF\ power}$

- Best-case: t = $\frac{5\ mJ}{5\ \mu W}$ = 1000s ≈ 17minutes

- Typical case: t = $\frac{5\ mJ}{1\ \mu W}$ = 5000s ≈ 1.4 hours

- Poor RF/Shielded bin: t = $\frac{5\ mJ}{0.2\mu W}$ = 25000s ≈ 7 hours

- Worst case/metal enclosure*: t = $\frac{5\ mJ}{0.05\ \mu W}$ ≈ 28hours

*Metal enclosure bins will have poor RF harvesting due to the skin effect.

Bin Node Radio Communication

- ❑Each bin integrates an **ultra-low-power sensing SoC**
- ❑Wireless communication uses **MRF24J40MA radio module**
- ❑Complies with **IEEE 802.15.4 standard (2.4 GHz ISM band)**
- ❑Designed for **short, low-data-rate, low-energy transmissions**
- ❑Enables **battery-less / energy-harvested bin operation**

Configuring the ESP32 Gateway

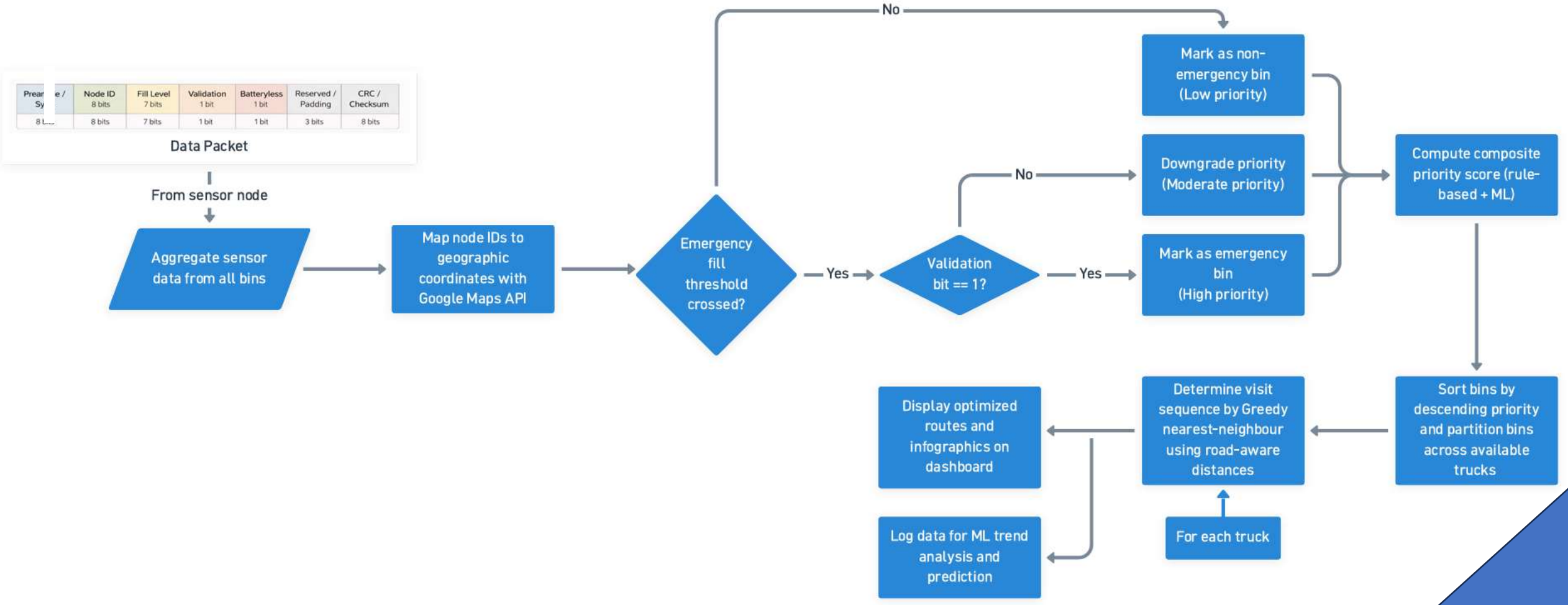
- ESP32 is configured as a **local IoT gateway**, deployed **one per locality**
- Powered using **mains / USB**, enabling **continuous operation**
- Connected to an **IEEE 802.15.4 radio module (MRF24J40MA)** via **SPI interface**
- Configured to **listen for sensor packets** from multiple bin nodes
- Received packets are **decoded and structured into JSON format**
- JSON data is transmitted to the cloud using **MQTT over Wi-Fi**

- The ESP32 gateway uses the **ESP32 Wi-Fi library** for internet connectivity, **SPI** library with an **MRF24J40 IEEE 802.15.4 driver** for radio communication, **ArduinoJson** library for data structuring, and **PubSubClient** library for MQTT-based cloud transmission.

Cloud Configuration

- ❑Choice of cloud service provider: AWS
- ❑Hosts business logic for priority scheduling and route optimization
- ❑Hosts ML inferencing for trend analysis and future predictions
- ❑Data from all localities is aggregated at the **cloud layer** via ESP32 gateways.

Cloud Business Logic



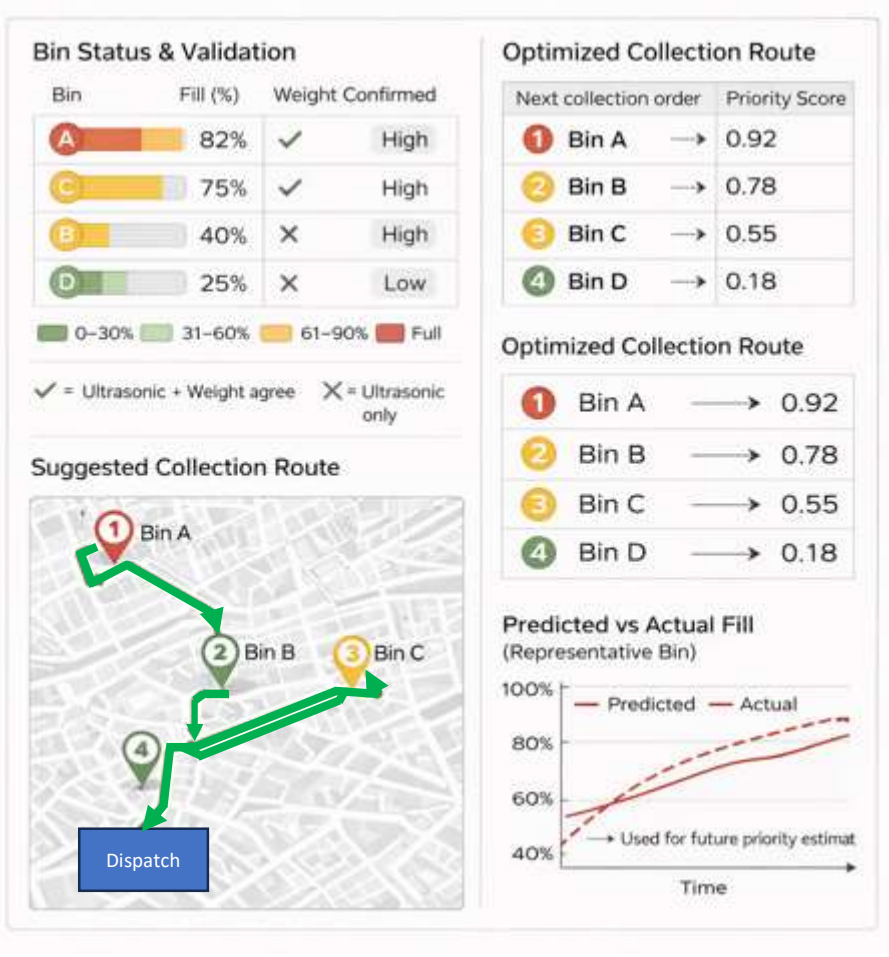
City Scale priority sorting (Hosted on Cloud)

- ❑Sensor data (fill level, validation bit, ID of each node corresponds to their physical location) from all bins is aggregated at the **cloud layer**
- ❑Bins exceeding the **emergency fill threshold** are shortlisted (**fill level = highest priority**)
- ❑The **validation bit** confirms true overflow and filters false positives (**second priority**)
- ❑**Distance from the nearest dispatch center** is evaluated to minimize operational cost
- ❑A **lightweight ML model (Logistic Regression / Gradient-Boosted Trees)** generates a final urgency score
- ❑Bins are given **composite priority score** and sorted in **descending order** for collecting

Route Optimization Strategy (Hosted on Cloud)

- **Google Maps Geocoding API** is used at the cloud layer to obtain accurate **latitude–longitude coordinates** for bins and truck dispatch centres
- Emergency bins are first **priority-sorted**, and the city is partitioned into **multiple routes based on available trucks**
- For each truck, the **dispatch centre is treated as the start node**, and bins assigned to that truck form the routing set
- A **greedy nearest-neighbour algorithm** combined with **Google Maps distance data** (road-aware distances) determines the visit sequence
- The system generates **multiple optimized routes simultaneously**, minimizing travel distance while ensuring high-priority bins are serviced first

Dashboard



- This is a conceptual dashboard for system monitoring, not a final UI.

Cost and Budget Analysis

Category	Assumption (GHMC-Based)	Estimated Value
City Administration	Greater Hyderabad Municipal Corporation (GHMC)	~150 wards (localities)
Smart Bins per Locality	Average of dense + residential areas	~50 bins / ward
Total Smart Bins	150 wards × 50 bins	**~7,500 bins**
Gateways	1 ESP32 gateway per locality	**150 gateways**
Cost per Smart Bin	PIC24 + sensors + radio (low-power IoT)	**₹1,500 / bin**
Cost per Gateway	ESP32 + radio module	**₹1,200 / gateway**
Total Bin Hardware Cost	7,500 × ₹1,500	**₹1.125 crore**
Total Gateway Cost	150 × ₹1,200	**₹0.018 crore**
Total One-Time Hardware Cost	Bins + gateways	**≈ ₹1.14 crore**
Cloud + APIs (Annual)	AWS + MQTT + Maps + ML (city-wide)	**≈ ₹0.0023 crore/year**
Cost per Locality	50 bins + 1 gateway	**≈ ₹76,000 / ward**

Summary

- Proposed a **battery-less, RF-energy-harvested smart waste monitoring system** enabling maintenance-free operation.
- Designed a **low-power sensor node** combining ultrasonic fill sensing with mechanical validation to reduce false positives.
- Implemented **event-driven communication** using IEEE 802.15.4, minimizing transmission energy.
- Developed a **cloud-based decision framework** that prioritizes bins using fill level, validation, distance, and ML-based urgency scoring.
- Demonstrated **scalable multi-truck route optimization** using greedy, road-aware routing.
- Achieved an **end-to-end, deployable architecture** spanning edge hardware, gateway, cloud intelligence, and visualization.

Future Work

- **Hybrid energy harvesting:** Combine RF harvesting with solar or vibration energy to increase wake-up frequency and reliability.
- **Custom integrated SoC:** Replace discrete components with a single ultra-low-power ASIC for further size and energy reduction.
- **Advanced routing algorithms:** Incorporate capacity constraints, time windows, and dynamic traffic-aware routing.
- **Predictive scheduling:** Use long-term ML models to schedule proactive collections before emergency thresholds are reached.
- **Field deployment & validation:** Conduct real-world trials to measure RF availability, packet loss, and system robustness.
- **Security extensions:** Add lightweight authentication and encryption suitable for energy-constrained nodes.

Thank You!